

National Academy of Science and Technology
Dhangadhi, Kailali
Pre-University Examinations
Semester: V_Fall

Level: Bachelor
Program: B.E. Computer
Course: Probability and Statistics

Year : 2025
F.M : 100
P.M : 45
Time : 3Hrs.

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

- 1 a.) Draw less than and more than an ogive curve of the following frequency distribution:

Marks	10-20	20-30	30-40	40-50	50-60	60-70
No. of students	4	6	10	20	18	2

Also locate the median value.

[7]

- b) The lives of two models (A and B) of refrigerators in a recent survey are shown below:

Life (No. of Years)	No. of refrigerators	
	Model A	Model B
0-2	5	2
2-4	16	7
4-6	13	12
6-8	7	19
8-10	5	9
10-12	4	1

- i) What is the average life of each model of these refrigerators?
ii) Which model has greater uniformity?

[8]

- 2.a) The weights (in grams) of 10 orange picked at random from a basket are as Follows: 45, 30, 75, 40, 60, 65, 70, 85, 80, 90

[7]

- i. List the five number summary.
ii. Form the box and whisker plot and describe the shape.
b) State and prove Baye's theorem.

[8]

OR

A factory has 3 units A, B and C. A produces 25% of its products, unit B produces 25% and the unit C produces 50%. If the percentage of

defective items produced by three units A, B and C respectively 1%, 2% and 3% and an item selected randomly from the total production of the factory is found to be defective. What is the probability that it is produced by the unit C?

3.a) If 5% of the electric bulbs manufactured by a company are defective, use poisson distribution to find the probability that in a sample of 100 bulbs,

- none is defective
- at least one is defective
- at most 2 bulbs are defective
- 5 bulbs are defective

b) In an examination 15% of the students got first class 60 marks above, while 40% securing below 40 marks. Assuming the marks are normally distributed, estimate mean and standard deviation.

4.a) In a certain city, the daily consumption of electric power (In millions of Kilowatt-hours) can be treated as a random variable having a gamma distribution with $\alpha = 3$ and $\beta = 2$. If the power plant of this city has a daily capacity of 12 million of kilowatt hours. What is the mean of this gamma distribution? What is the probability that this power supply will be inadequate on any given day?

b) If $f(x,y) = 4xy$, $0 < x < 1$, $0 < y < 1$
 $= 0$, otherwise

- Verify that $f(x,y)$ is a p.d.f.
- Find the marginal distribution of X and Y.
- Find the conditional distribution of X and Y.
- Find whether X and Y are independent or not.

5.a) A random sample of 10 balls have the following life in month : 24, 26, 32, 28, 20, 20, 23, 34, 30 and 43. Obtain 95% confidence limit for the population mean.

b) Before an increase in excise duty on coffee 80 out of sample of 100 persons were known to be taking coffee. After the increase in duty 60 persons are now found to be taking coffee in a sample of 120. Do you think that there has been a significant decrease in the consumption of

coffee after the increase in the excise duty? Test at 1% level of significance?

6.a) An I.Q. test was administered to 5 persons before they were trained. The results are given below

Persons	1	2	3	4	5
IQ before training	110	120	123	132	125
IQ after training	120	118	125	136	121

Test whether there is any effective in IQ after the training program.

OR

A random sample of 90 adults are classified according to gender and the number of hours they watch television during a week.

Hours	Male	Female
Over 25 years	15	29
Under 25 years	27	19

Test the hypothesis that time spent on watching television is Independent of whether viewer is male or female at 5% level of significance.

b) The following are the heights in centimeter and weights in kilogram of 8 men:

Height	160	168	174	176	180	181	182	185
Weight	65	66	68	70	75	76	78	80

- Develop the estimating regression equation of weight on height.
- Estimate the weight of men whose height is 175cm.

7. Write short notes on (any two):

- Source of Data.
- Characteristics of normal distribution
- Criteria of good estimator